

The Effects of State Minimum Wage Increases on Restaurant Employment and Prices

Evidence from California and New York, 2010–2023

Anjeela Budha Magar

Econ 5253 Spring 2026

Section 1

Introduction

What this paper asks

The minimum wage debate has run for thirty years. Two big questions remain.

1. **Employment.** Do large state minimum wage increases reduce restaurant jobs, as the textbook model predicts, or not, as *Card and Krueger (1994)* argued?
2. **Prices.** If firms don't lay workers off, do they pass the cost on to consumers through higher menu prices?

Setting. California and New York both began phasing in \$15-per-hour minimum wages in 2014. These are among the largest state-level wage shocks in U.S. history — a sharp natural experiment.

What I do. Two parallel difference-in-differences models, one for each treated state, on annual food-services employment and food-away-from-home CPI, 2010–2023.

Section 2

Data

Data sources and state pairings

All series come from FRED.

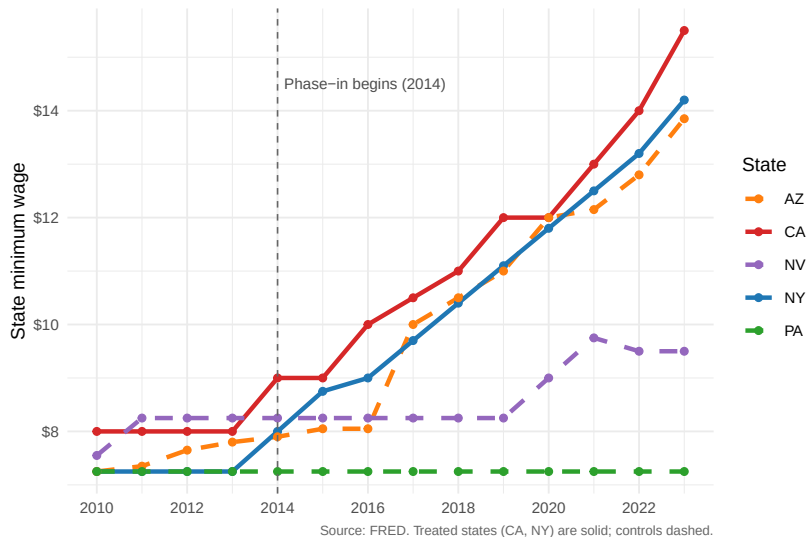
Table 1: Data inputs and aggregation.

Variable	Source	Frequency	Annualized by
State minimum wage	STTMINWG{CA,NY,AZ,NV}	Annual	January value
Real GSP	{CA,NY,AZ,NV,PA}RQGSP	Quarterly	Annual mean
Food-services employment	SMU??000007072200001SA	Monthly	Annual mean
CPI: food away from home	CUURA{422,101,102}SEFV	Monthly	Annual mean
CPI: Phoenix-Mesa (AZ proxy)	CUUSA429SEFV	Annual	(none)

Two state pairings:

- **California** (treated) vs **Arizona + Nevada** (control) — shared sunbelt geography, tourism-heavy hospitality, similar seasonal patterns. AZ and NV moved their wage floors much more slowly than CA after 2014.
- **New York** (treated) vs **Pennsylvania** (control) — long shared border, integrated commuter labor markets, similar Mid-Atlantic industrial mix. PA stayed at the federal floor of \$7.25 throughout.

Minimum wage trajectories



Section 3

Methods

Difference-in-differences specification

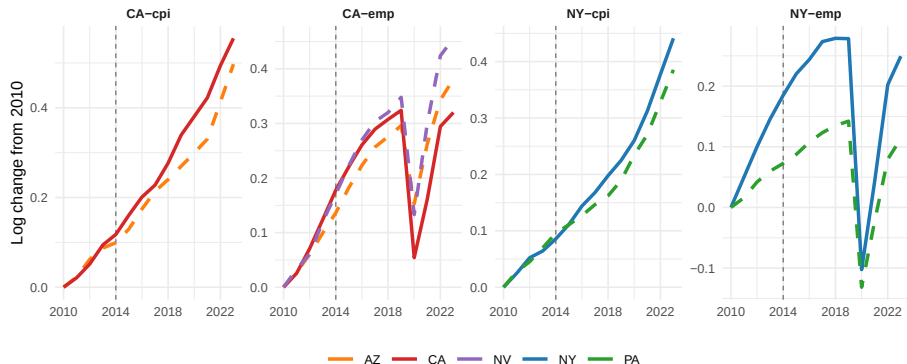
Estimating equation:

$$\ln y_{st} = \alpha + \beta \cdot (\text{Treat}_s \times \text{Post}_t) + \gamma_s + \delta_t + X'_{st}\pi + \varepsilon_{st}$$

- $\ln y_{st}$ = log food-services employment or log food-away-from-home CPI in state s , year t .
- $\text{Treat}_s = 1$ for CA (or NY); $\text{Post}_t = 1$ for years ≥ 2014 .
- γ_s, δ_t = state and year fixed effects.
- X_{st} = log real GSP (state business cycle); for employment I also test specifications with the one-year lag of $\ln y$.
- β = average treatment effect on the treated under parallel trends. Standard errors are HC1 robust.

Identifying assumption: treated and control states would have followed parallel trajectories absent the policy. I check this with pre-2014 growth rates next slide.

Pre-trends check



Solid = treated state. Dashed vertical line = 2014 phase-in.

Read: CA, CA-cpi, and NY-cpi panels track their controls closely before 2014 — parallel-trends assumption looks credible. The **NY-emp** panel does not: New York food-services employment was already growing about 2.9 pp/yr faster than Pennsylvania pre-2014. NY-emp results need to be read with that caveat.

Section 4

Findings

Main DiD results

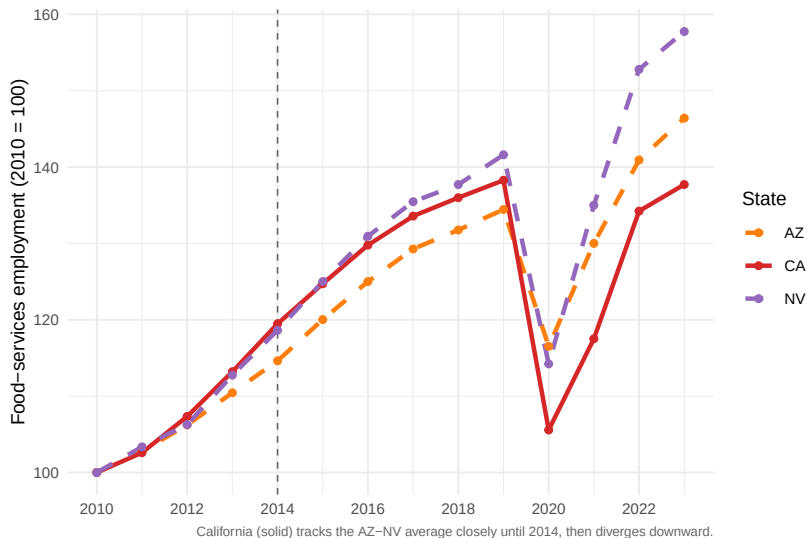
Table 2: DiD coefficient on (Treat $\times$ Post).

Specification	DiD coef.	% effect	SE (HC1)	p-value	N	R ²
Employment outcomes (log food-services employment)						
CA-emp: no controls	-0.0369	-3.62%	0.0182	0.053	42	0.999
CA-emp: + ln(GSP)	-0.0641	-6.21%	0.0236	0.012	42	0.999
CA-emp: + ln(GSP) + lag	-0.0209	-2.07%	0.0275	0.454	39	1.000
NY-emp: no controls	+0.0719	+7.46%	0.0212	0.005	28	0.995
NY-emp: + ln(GSP)	+0.0892	+9.33%	0.0196	0.001	28	0.996
NY-emp: + ln(GSP) + lag	+0.0474	+4.85%	0.0309	0.160	26	0.997
Price outcomes (log CPI food-away-from-home)						
CA-cpi: no controls (AZ only)	+0.0521	+5.35%	0.0099	0.000	28	0.999
CA-cpi: + ln(GSP)	+0.0693	+7.17%	0.0135	0.000	28	0.999
NY-cpi: no controls	+0.0276	+2.80%	0.0072	0.002	28	0.998
NY-cpi: + ln(GSP)	+0.0119	+1.20%	0.0059	0.069	28	0.999

Three patterns to highlight:

- **CA employment** ↓ 2–6% relative to AZ+NV — the textbook prediction shows up here.
- **CA prices** ↑ 5–7% relative to AZ — substantial pass-through of wage costs to menu prices.

Visual evidence: California employment



Section 5

Conclusion

What did we learn?

Three takeaways for California:

- Modest **disemployment** (-2 to -6%) in food services, in line with the recent literature on large wage shocks (*Jardim et al. 2022, Seattle*).
- Substantial **price pass-through** ($+5$ to $+7\%$) — consistent with *Aaronson (2001)* and *Harasztosi & Lindner (2019)*: firms absorb wage shocks mostly through prices, partly through employment.
- Net effect on workers is ambiguous: those who keep their jobs earn more; those displaced lose; consumers pay more.

Why it matters:

- Federal and state minimum wage debates ($\$15+$ proposals nationwide) need credible estimates of *both* employment and price effects.
- Single-outcome studies miss half the story — a paper that looks only at jobs would conclude “not much happened”; one that looks only at prices would conclude “consumers paid for it.” The truth is both.

Honest caveats: small number of states (low power), state-aggregate data masks within-state variation (NYC vs upstate), NY pre-trends fail the parallel test.